TripNest AI: Test Plan

Version 2.0 · 2 November 2026. Approach: tests are written first for each behaviour, then the code; real browser tests cover the user flow; failures are traced to their cause before fixing.

1. Scope

In: location handling, popular places, hidden gems, time-of-day timeline, best-time rules, crowd prediction, recommendation scoring, map data, stored photographs, security headers, configuration, responsive behaviour. Out (not part of the product): itineraries, community, reviews, weather, accounts, admin. Their old tests are kept in `backend/tests/legacy/` and are **not run**.

2. Levels and tools

Level	Tool	Location	What it proves
Unit and service	pytest	`test_crowd_engine.py`, `test_explore.py` (timeline, best window)	levels, simulator, chronological split, model beats baselines, timeline and best-time rules
API and integration	pytest + FastAPI test client	`test_core_features.py`, `test_explore.py`, `test_catalog_crowd.py`, `test_map_photos.py`, `test_schema_upgrade.py`	locations, explore, prediction, recommendations, map data, validation, removed modules not exposed, headers, health, schema upgrade
Browser end to end	Playwright in real Chrome, own server and database	`e2e/test_journeys.py`	the nine-step flow, denied and allowed location, date and time changes, map filters and popups, stale-explanation rule, photo failure, four screen sizes
Design checks	design-taste inspector and lint	reports not committed	layout measurements at 1440, 768 and 390 px
Documents	`docs/build_pdfs.py`	`docs/`	MRD is one A4 page, PRD two, PDFs open and have text, every cited test exists

3. Required scenario (nine steps)

Step	Test
1 Open the app	`test_step1_homepage_explains_the_product_and_offers_a_location_first_search`
2-3 Location permission or manual city, dynamic greeting	`test_denied_location_falls_back_to_a_typed_city`, `test_granted_location_is_used_with_a_dynamic_greeting`, `test_a_location_between_cities_is_described_as_near_one`, `test_steps2to4_typed_city_gives_a_dynamic_greeting_and_popular_places_with_predicted_crowds`
4 Popular places and predicted crowds for the date	`test_steps2to4_typed_city_gives_a_dynamic_greeting_and_popular_places_with_predicted_crowds` `test_changing_date_time_and_interest_refreshes_the_predictions`
5 Best time to visit	`test_step5_best_time_panel_shows_the_calmest_period_and_updates_per_destination`
6 Hidden gems	`test_step6_hidden_gems_are_lesser_known_calm_and_explained`
7 Interactive map	`test_step7_interactive_map_markers_popup_and_filters`, `test_view_on_map_scrolls_to_the_map_and_opens_that_place`
8-9 Open a destination and its prediction	`test_steps8and9_open_a_destination_and_see_its_prediction`, `test_destination_time_range_and_day_controls_update_the_prediction`
Screen sizes	`test_every_screen_size_supports_the_whole_flow` (1440, 1280, 768, 390 px)

4. Entry and exit criteria

Entry: dependencies installed from the pinned files; Chrome present; internet access for the browser tests (Leaflet, map tiles and photographs come from external hosts). Exit: all tests pass with no browser console or server errors; design inspector shows no unexplained failure; documents build with correct page counts; the deployment checklist passes on the live URL (**pending**).

5. Not covered (stated honestly)

Load testing; PostgreSQL; the Docker image build; the live deployment; browsers other than Chrome and real phones (phone testing is emulation); an accessibility audit with assistive technology; a dependency vulnerability scan; accuracy on real crowd data; the correctness of every hour, fee and coordinate in the catalogue (indicative public knowledge, positions approximate).